

Circuit-Switched Network

A circuit-switched network is a type of telecommunications network in which a dedicated communication path (circuit) is established between two endpoints before any data transmission takes place. Once the circuit is established, the communication path remains fixed and dedicated to the specific communication session until it is terminated.

In a circuit-switched network:

- **Dedicated Path:** A dedicated physical or virtual path is created between the sender and the receiver before data transmission. This path is reserved for the exclusive use of the communication session.
- **Fixed Bandwidth:** The dedicated circuit ensures that a fixed amount of bandwidth is allocated for the entire duration of the communication session, regardless of whether data is being actively transmitted or not.
- **Connection-Oriented:** Circuit-switched networks are connection-oriented, meaning that a connection must be established before data can be exchanged between the sender and receiver.
- **Real-Time Communication:** Circuit-switched networks are commonly used for real-time communication, such as voice calls or video conferencing, where a continuous and uninterrupted connection is required.
- **Predictable Delay and Quality:** Since the bandwidth is fixed and dedicated for the communication session, the delay and quality of the transmission are relatively predictable and consistent.

- **Resources Reservation:** Circuit-switched networks require resources to be reserved along the entire path of the circuit, even if data is not being transmitted, which can lead to inefficiencies in resource utilization.

Traditional telephone networks, known as Public Switched Telephone Networks (PSTNs), are examples of circuit-switched networks. When you make a phone call, a dedicated circuit is established between your phone and the recipient's phone, ensuring a continuous and dedicated connection during the call.

While circuit-switched networks provide stable and reliable connections, they are less efficient for transmitting bursty or sporadic data traffic compared to packet-switched networks. Packet-switched networks, like the modern Internet, use data packets that are routed independently, dynamically, and may share network resources, providing more efficient use of bandwidth. However, circuit-switched networks are still used for certain applications, especially for real-time voice and video communications.

Packet-Switched Network

A packet-switched network is a type of telecommunications network in which data is transmitted in the form of packets. Unlike circuit-switched networks, where a dedicated communication path is established before data transmission, packet-switched networks break data into smaller packets that are individually routed and transmitted over the network. These packets can take different paths to reach their destination and may be reassembled in the correct order upon arrival.

In a packet-switched network:

- **Packetization:** Data is divided into smaller packets before transmission. Each packet typically contains a portion of the data, along with header information, such as the source and destination addresses.
- **Dynamic Routing:** Packets are routed independently based on the destination address and the current network conditions. This means that different packets from the same data transmission may take different routes to reach their destination.
- **Store-and-Forward:** Each intermediate network node (routers or switches) receives the entire packet before it is forwarded to the next node. The packet is stored briefly for inspection and possible error correction before being forwarded.
- **Shared Resources:** In a packet-switched network, multiple packets from different data transmissions can share the same network resources (such as links and bandwidth), leading to more efficient utilization of network resources.

- **Variable Transmission Delay:** As packets may take different routes and experience varying congestion levels, the transmission delay for each packet can vary. This introduces some level of unpredictability in the packet delivery time.
- **Internet Protocol (IP):** The most common protocol used in packet-switched networks is the Internet Protocol (IP), which allows data to be routed across interconnected networks, including the global Internet.

Packet-switched networks, like the modern Internet, have become ubiquitous due to their flexibility, scalability, and efficiency. They can handle a wide range of data types, including text, images, audio, and video, making them suitable for various applications like web browsing, file transfer, email, video streaming, and online gaming.

The data packets are assembled at the destination based on the information in their headers, ensuring that the original data is reconstructed correctly. Packet-switched networks allow for efficient use of network resources and are well-suited for bursty or sporadic data traffic, where data is sent in irregular intervals.

Virtual Circuit Network

A virtual circuit network is a type of telecommunications network that provides a connection-oriented communication service between two endpoints. In a virtual circuit network, a logical communication path, called a "virtual circuit," is established between the sender and receiver before data transmission occurs. This virtual circuit acts as a dedicated and pre-configured pathway for the entire communication session.

Key characteristics of a virtual circuit network:

- **Connection-Oriented:** Virtual circuit networks are connection-oriented, meaning that a dedicated logical connection is established between the communicating parties before data transfer begins. The setup phase involves reserving network resources and configuring the network for the specific communication session.
- **Virtual Circuit Establishment:** During the setup phase, a virtual circuit is created with a unique identifier (such as a Virtual Circuit Identifier or VCI) that helps the network devices identify and route packets belonging to that specific virtual circuit.
- **Fixed Routing:** Once the virtual circuit is established, all packets belonging to the same communication session follow the same predetermined route through the network. The routing decisions are made during the setup phase and remain fixed throughout the session.

- **Reliable Delivery:** Virtual circuit networks provide reliable delivery of packets along the established path. If a packet is lost or corrupted, the network is responsible for retransmitting it or handling error recovery.
- **Low Overhead:** Since the virtual circuit is established beforehand, there is less overhead required for addressing and routing each packet during data transmission.
- **Guaranteed Order:** Packets within the virtual circuit are delivered in the same order as they were sent, ensuring that the data is reconstructed correctly at the receiving end.
- **Tear-Down Phase:** Once the communication session is complete, the virtual circuit is terminated, and the network resources are released for other uses.

Virtual circuit networks are commonly used in Asynchronous Transfer Mode (ATM) and Frame Relay networks. ATM networks use Virtual Path Identifier (VPI) and Virtual Circuit Identifier (VCI) to identify virtual circuits, while Frame Relay networks use Data Link Connection Identifier (DLCI) for the same purpose.

Compared to packet-switched networks, like the modern Internet, virtual circuit networks may have higher setup and tear-down overhead. However, they are well-suited for applications that require guaranteed delivery, fixed routing, and predictable network behavior, such as real-time voice and video communication.

Message Switched Network

Message switching is a type of telecommunications network that predates both circuit-switched and packet-switched networks. In a message-switched network, data is transmitted in the form of complete messages, where a message consists of the entire data unit to be sent, along with destination address and other control information. The entire message is stored and forwarded through the network from one node to another until it reaches its destination.

Message switching was used in early telecommunication systems, such as telex networks, where text-based messages were transmitted between teleprinters. As technology evolved, message switching was largely replaced by more efficient circuit-switched and packet-switched networks.

Today, packet-switched networks, like the Internet, dominate data transmission due to their flexibility, scalability, and efficient use of network resources. However, the concepts of message switching have contributed to the development of modern networking protocols and techniques.